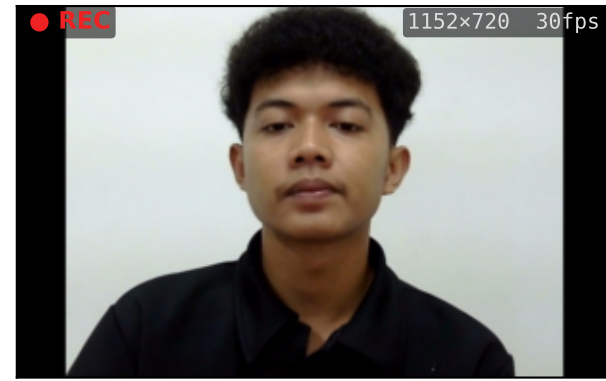


Data Preprocessing Pipeline — Primer dataset (frontonly_conf60)

(1) Raw video frame



frame dari
rekaman webcam

(2) Face detection



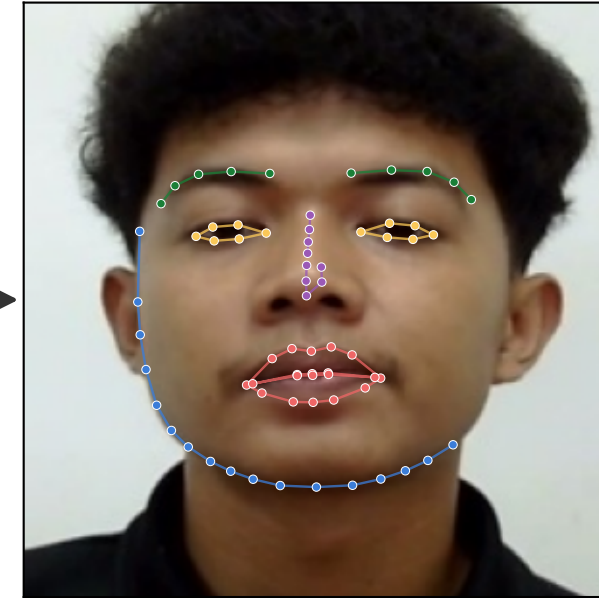
MediaPipe FaceLandmarker
bounding box

(3) Crop 224x224



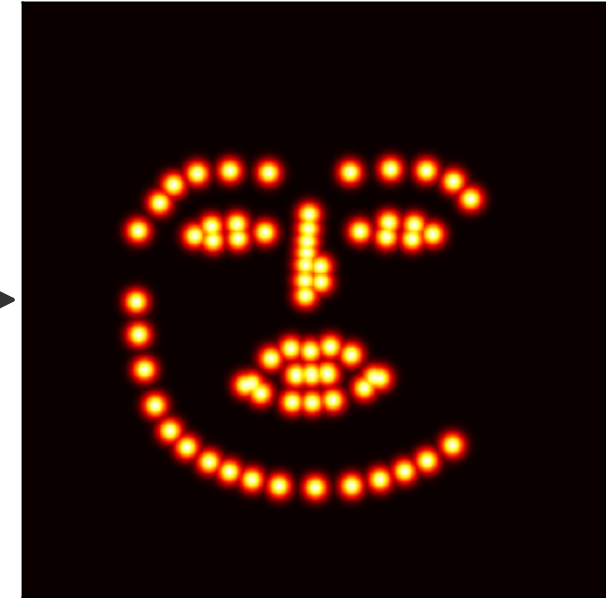
face crop
resize ke 224x224

(4) Landmark extraction



68 titik MediaPipe (MP)
(FA di-extract paralel)

(5) Heatmap rendering



Gaussian blob
per landmark → channel-4

OUTPUT yang disimpan ke disk per sample:

- RGB image (224x224x3) → input CNN / image branch
- Landmark raw_136 (68x2) — 2 source paralel: face-api.js (FA) + MediaPipe (MP)
→ derive ke FACS_28 (Euclidean dist), Blendshape_52 (MP only), atau FACS+BS_80
- Heatmap (224x224) → channel ke-4 input Early Fusion